- **Graph vs embedding**: graphs are abstract data structures; to visualize them, it's possible to *embed* a graph to a geometrical space; for example, it's possible to create embeddings to the plane.
- **Plane graph**: an embedding is a mapping between vertices and points in an Euclidean space, and between edges and (Jordan) curves in the same space.
- **Planar embedding**: If an embedding maps a graph to the plane, and none of the edges intersect another edge or a vertex (besides its endpoints), then the embedding is called a planar embedding. A graph that has a planar embedding is called a planar graph.
- **Crossing number**: While it's possible to embed all graphs in the 3D Euclidean space without edge intersections, not all graphs are planar. For a non-planar graph `G`, we define its *crossing number* as the minimum number of edge intersections that any embedding of `G` to the plane can have. Planarity testing can be performed efficiently, but instead finding the crossing number of a generic graph is an NP-complete problem.
- **Rectilinear crossing number**: If we restrict the shape of the edges to straight-line segments, instead of generic Jordan curves, for some graphs it's not possible to find a *straight-line drawing* with the minimum possible number of intersections. We define the *rectilinear crossing number* of a graph `G` as the minimum number of intersections of all embeddings of `G` where edges are drawn using straight-line segments.
- **Edge intersection**: computing if and how edges cross is not easy; in straight-line drawings it's less expensive because checking two segments only requires a constant number of operations on vectors; when